

Mohammad Mahmudul Hasan

Geospatial AI and GIS | Team Lead, AI Engineer

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RESEARCH PROFILE

AI engineering team lead with professional experience in orthophoto workflows, GIS digitization, multi-year building-change detection, point-cloud processing and road-asset inspection. Seeking graduate research in GeoAI, urban mapping and robust change analysis.

EDUCATION

Islamic University of Technology, Bangladesh | BSc in Electrical and Electronic Engineering, completed 22 March 2021.

CGPA 3.76/4.00; first class with honours. Project and Thesis: A+; Digital Signal Processing: A; Artificial Neural Networks and Fuzzy Logic: A+.

PROFESSIONAL EXPERIENCE

HawalIT Limited — Team Lead, AI Engineer | with the company since 2021; team-lead role since [ADD month and year of promotion].

Currently lead a six-member AI engineering team.

- Worked on multi-year orthophoto comparison for building construction, demolition and extension analysis.
- Worked on GIS digitization from orthophotos and traffic-sign condition or missing-asset review using car-captured 360-degree imagery.
- Processed and interpreted raster and point-cloud data across production mapping tasks; currently lead a six-member AI engineering team.

Earlier titles, exact dates and individual project ownership: [ADD verified career chronology].

PEER-REVIEWED PUBLICATION

Tasnim Sakib Apon, Mohammad Mahmudul Hasan, Abrar Islam, and Md. Golam Rabiul Alam. “Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI.” 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), 2021. DOI: [10.1109/CSDE53843.2021.9718400](https://doi.org/10.1109/CSDE53843.2021.9718400).

Individual contribution: [ADD precisely what you did]. The author preprint is the same work.

SELECTED RESEARCH AND TECHNICAL WORK

BUILDING CHANGE ACROSS ACQUISITION YEARS | Professional experience

Worked with multi-year orthophotos to identify building construction, demolition and extension.

Research question: How can geometric checks and temporal reasoning reduce false change detections across surveys?

GIS DIGITIZATION FROM ORTHOPHOTOS | Professional experience

Worked on digitization using orthophoto products, connecting image interpretation with mapped features.

Research question: How can AI-assisted extraction reduce review effort while maintaining geometric and topological quality?

TRAFFIC-SIGN REVIEW FROM 360-DEGREE IMAGERY | Professional experience

Worked with car-captured 360-degree imagery to identify faulty or missing traffic signs and review their quality.

Research question: Can a visual system distinguish a genuine defect from an uncertain observation and communicate the evidence to a reviewer?

RESEARCH DIRECTION

How can GeoAI distinguish real changes in the built environment from differences introduced by acquisition and processing? Study building-change detection with explicit registration checks and spatially separated evaluation. Connect model outputs to usable map updates and measure both detection quality and review effort.

TECHNICAL FOUNDATION

GIS digitization; geospatial imagery; building-change analysis; point-cloud classification; data preparation; Python; MATLAB. GIS software, spatial databases and ML frameworks: [ADD verified tools].

ADDITIONAL CONTEXT

Independent AI/ML study during the final semesters of the EEE degree and participation in AI competitions. Historical Coursera credentials and competition placement should be added with year and verification links.

Language: Bangla (native); English-medium instruction certificate available. IELTS pending; intended overall score 7.0 is not an achieved score.

References: [ADD academic and professional referees as required by the program].